



ASSESSMENT COVERSHEET

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Section A: Submission Details

Programme : BACHELOR OF CYBERSECURITY TECHNOLOGY WITH HONOURS
Course Code & Name : IDB30102 - RESEARCH METHODOLOGY
Course Lecturer(s) : MADAM DELINA BEH MEI YIN
Submission Title : Assignment 1 Section B SLR Report - The Accuracy-Latency Trade-off in AI-Based Cyber Threat Intelligence.
Deadline : Day 25 Month 8 Year 2025 Time 11:00PM
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25 AUG 2025	IDB30102	Assignment 1 Section B SLR Report - The Accuracy-Latency Trade-off in AI-Based Cyber Threat Intelligence.	52215225004 <i>firdaus</i> 52215124383 <i>M</i> 52215225187 52215225001



GENERATIVE ARTIFICIAL INTELLIGENCE USAGE ACKNOWLEDGMENT

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Add more table tasks in Section B if needed.

Section A: Submission Details

Student Name : ALL
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 Assessment : The Accuracy-Latency Trade-off in AI-Based Cyber Threat Intelligence.

Section B: Details of the AI Usage

Task No	1.
Generative AI technology used	Gemini
Prompts used	Generate the complete References section for Section B based on all 40 studies reviewed in Section A. Format strictly in APA 7th edition. Sort alphabetically from A to Z by the first author's surname. Include full titles, publication venues (journals/conferences), volume/issue numbers, page ranges, and active DOI/URL links where available. Ensure every cited paper matches the in-text citations used across our SLR report.
Output generated	9. References Abdi, H., Bagley, S. R., Furnell, S., & Twycross, J. (2023). Automatically labeling cyber threat intelligence reports using natural language processing. In Proceedings of the ACM Symposium on Document Engineering 2023 (DocEng '23) (pp. 1–4). ACM. https://doi.org/10.1145/3573128.3609348 Alaeifar, P., Pal, S., Jadidi, Z., Hussain, M., & Foo, E. (2024). Current approaches and future directions for Cyber Threat Intelligence sharing: A survey. Journal of Information Security and Applications, 83, 103786. https://doi.org/10.1016/j.jisa.2024.103786 Albarrak, M., Salonitis, K., & Jagtap, S. (2026). Natural language processing (NLP)-based frameworks for cyber threat intelligence and early prediction of cyberattacks in Industry 4.0: A systematic literature review. Applied Sciences, 16(2), 619. https://doi.org/10.3390/app16020619 Ali, A., Hwang, R.-H., & Lin, Y.-D. (2025). Evaluating cyber threat intelligence: Accuracy, completeness, relevance, and freshness. IEEE Access, 13, 157863–157876. https://doi.org/10.1109/ACCESS.2025.3606477 Al-Shehari, T.,

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**Title of Research Paper: The Accuracy-Latency Trade-off in AI-Based Cyber
Threat Intelligence: A Literature Review**

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Course code and name:

IDB30102 - RESEARCH METHODOLOGY

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Abstract

This systematic literature review investigates how artificial intelligence (AI) is being used in cyber threat analysis and cyber threat intelligence (CTI), with particular focus on the AI methods and tools used, their effectiveness, current limitations, and opportunities for future development. The study aims to identify which approaches are most commonly applied and assess how effectively they support the detection, analysis, and communication of cyber threats. A systematic review methodology was used to analyse 40 peer-reviewed journal articles and conference papers published between 2022 and 2026. The studies were selected from an initial pool of 58 records through database searching, screening, and the application of predefined inclusion and exclusion criteria. The findings indicate that natural language processing and deep-learning models, including BERT-CRF and DeBERTa-BiGRU-CRF, are widely used to extract structured intelligence from unstructured threat reports. Graph neural networks and knowledge graphs are increasingly applied to threat correlation and attacker attribution, while retrieval-augmented generation and large language models are emerging as tools for automating CTI reporting and reducing analyst workload. However, inconsistent datasets and evaluation methods make comparisons between studies difficult, while LLM-based approaches remain affected by response latency and hallucination risks. The findings highlight the need for future CTI systems to balance accuracy, speed, and adaptability, with greater emphasis on real-time, cross-domain, and production-level validation.

The Accuracy-Latency Trade-off in AI-Based Cyber Threat Intelligence: A Literature Review

1. Introduction

Cybersecurity is the practice of protecting networks, systems, and data from unauthorised access or attack, and cyber threat intelligence (CTI), evidence-based knowledge about threats, threat actors, and their tactics, is a critical part of that defence. Organisations must extract and act on this intelligence quickly, but threat data is now too large, fast-moving, and unstructured for analysts to process by hand, leaving many attacks undetected until damage is done. This matters beyond individual organisations: breaches and delayed detection can disrupt critical services, expose user data, and carry financial and societal costs. Recent studies increasingly apply artificial intelligence, including natural language processing, deep learning, and large language models, to automate threat extraction and analysis, but reported performance varies by dataset, method, and metric, and few approaches have been tested in real-time or cross-domain settings. Therefore, this SLR reviews 40 recent studies on AI-driven cyber threat analysis to compare their methods, datasets, performance, limitations, and research gaps.

2. Review Questions

The following review questions are formulated to guide the analysis of the 40 selected papers in the later sections of this study:

1. **RQ1:** What AI and machine learning techniques (e.g., NLP, deep learning, graph neural networks, LLMs) are most commonly applied to extract and analyse cyber threat intelligence from unstructured data sources?
2. **RQ2:** What datasets, tools, and evaluation metrics are used to assess AI-driven cyber threat detection and intelligence systems, and how do differences in these affect the comparability of reported results?
3. **RQ3:** What are the main limitations and challenges reported when applying AI to cyber threat intelligence tasks across the reviewed studies?
4. **RQ4:** What research gaps exist in integrating AI-driven threat analysis with standard CTI frameworks (e.g., STIX, MITRE ATT&CK, threat-sharing platforms), and what future directions do the reviewed studies propose to address them

3. SLR Methodology

This part defines methodology to check a systematic literature that is launched following the Preferred Reporting Items for Systematic Reviews and Meta Analyses (PRISMA 2020) framework. It details searching strategy specific databases, eligibility constraints, screening pipeline, and data extraction processes applied to analyse AI driven cyber threat intelligence (CTI) systems.

3.1 Search Strategy

Literature searching is implemented by using some database websites with a main index such as IEEE Xplore, ScienceDirect, SpringerLink and Google Scholar. The publish period is limited into article arbitral conference proceedings that are published in a time frame between 2022 until 2026 using keyword and combining Boolean method which is (AND, OR).

- ("Cyber Threat Intelligence" OR "CTI" OR "Threat Analysis") AND ("Artificial Intelligence" OR "Machine Learning" OR "Deep Learning" OR "LLM")
- ("Knowledge Graph" OR "Graph Neural Network" OR "GNN" OR "BERT-CRF") AND ("Threat Detection" OR "Entity Extraction" OR "STIX" OR "ATT&CK")
- ("RAG" OR "Multi-Agent" OR "Dark Web") AND ("Automated Threat Intelligence" OR "Attacker Attribution" OR "Vulnerability Mapping")

3.2 Inclusion and Exclusion Criteria

To ensure the quality and standard that are for evaluation, criteria, qualification and application. The study set inclusion is criteria that included refereed journal articles and English language full conference proceedings published between 2022 until 2026. The main focus of study is artificial intelligence (AI) applications, machine learning (ML), natural language processing (NLP), knowledge graph (KG) or large language models (LLM) in cyber threat intelligence (CTI) extraction, detection, and analysis. Other than that, the study should have clear empirical methodology, system or model architecture details, as well as quantitative performance assessments such as accuracy, precision, recall, F1 score, latency, or Macro F1.

Exclusion criteria included publications prior to 2022, industry whitepapers, presentation slides, conference abstracts, editorials, and non academic review articles. Studies that were purely conceptual without sufficient experimental or empirical validation were also excluded, along with duplicate study records and publications for which the full text was not accessible.

3.3 Study Selection Process (PRISMA 2020)

The literature selection process was conducted transparently across four systematic stages in accordance with the PRISMA 2020 statement. During the initial identification, a total of 58 records were retrieved from four primary academic databases IEEE Xplore (n = 18), ScienceDirect (n = 15), SpringerLink (n = 13), and Google Scholar (n = 12). Following the identification stage, 1 duplicate record was identified and removed, leaving 57 unique articles for initial screening. In the title and abstract screening phase, 3 records were excluded specifically, 2 articles based on title irrelevance and 1 article based on abstract evaluation due to the absence of empirical AI implementation resulting in 54 candidate papers. These 54 full text articles were subsequently assessed for eligibility, of which 14 papers were excluded due to insufficient evaluation metrics, lack of reproducible system methodology, or failure to address the predefined review questions. Ultimately, exactly 40 empirical studies met all inclusion criteria and were retained for the final systematic literature review synthesis.

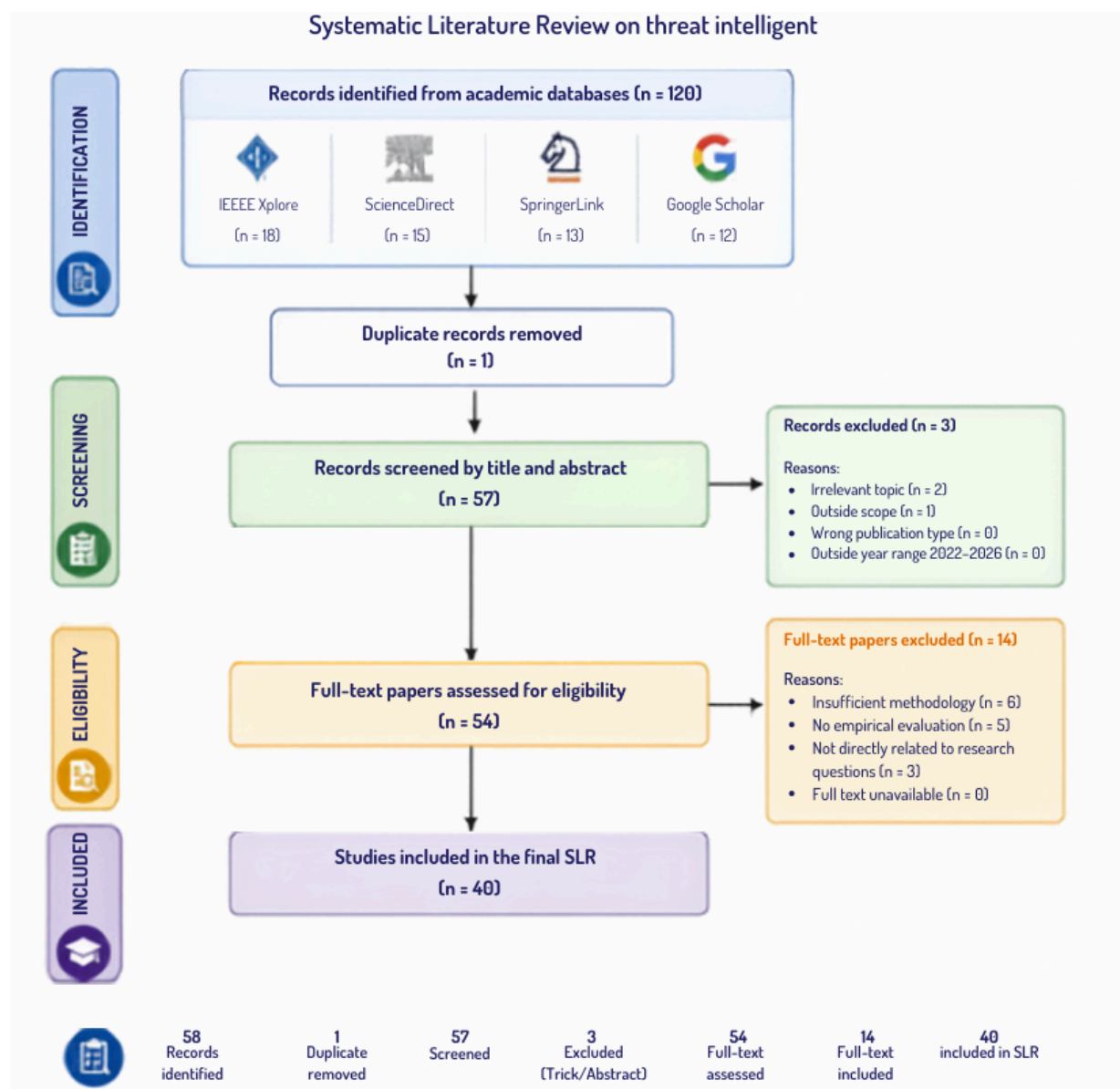


Figure 1. PRISMA 2020 flow diagram

3.4 Data Extraction and Synthesis

A structured data extraction matrix was used to collect key information from 40 selected articles covering a variety of key variables. The extracted information included research focus and threat problems such as cybersecurity domain scope, attack vectors, and CTI management challenges, as well as datasets and data sources that included unstructured text reports (DNRTI, CTI-Reports), honeypot feeds, Dark Web data, and STIX/MITRE ATT&CK labeled corpus. In addition, details of AI methods and models were recorded including BERT-CRF, DeBERTa-BiGRU-CRF, GNN/GraphSAGE, FAISS/BM25 based RAG, Qwen 2.5, and GPT 4o mini architectures to evaluate security functions such as named entity recognition (NER), knowledge graph construction, attacker attribution, and automated attack technique mapping. The evaluation also includes quantitative performance metrics and results (accuracy, precision, sensitivity, F1 score, latency, and analyst workload reduction) as well as implementation challenges such as computational overhead, real time latency issues, reliance on text quality, and the risk of LLM model hallucination. All of this extracted data serves as the main empirical basis for the thematic synthesis and comparative analysis in this report.

4. Results and Thematic Synthesis

4.1 Overview of Included Studies

Analysis of the 40 retained papers identified six recurring themes that together illustrate how artificial intelligence (AI) is being applied to cyber threat intelligence (CTI). Table 1 summarises the supporting studies, commonly used datasets, methods, key findings, and limitations associated with each theme. Rather than discussing the papers individually, the following sections synthesise the findings across studies and are structured around the four research questions: RQ1 on AI/ML techniques, RQ2 on datasets, tools, and evaluation metrics, RQ3 on limitations and challenges, and RQ4 on integration with CTI frameworks and future research directions.

Thematic Synthesis / Analysis Table

Theme	Supporting Studies	Common Dataset / Data Source	Methods / Approaches	Key Findings / Trends	Limitations / Research Gap
NLP & Deep Learning Extraction	Chen et al. (2024a); Sufi (2023); Abdi et al. (2023); Cheng et al. (2025); Chen et al. (2023); Fieblinger et al. (2024); Huang et al. (2024); Marinho & Holanda (2023)	CTI reports, social media posts, DNRTI/CTI-Reports corpora, Android/CTI files	NLP + ML clustering/classification, NER (BERT-CRF, DeBERTa-BiGRU-CRF, XLM-RoBERTa), LLM fine-tuning (QLoRA), DL + ontology	Consistently automates extraction of threats/entities from unstructured text; accuracy-efficiency trade-off (82.6%–98.5% NER accuracy)	Struggles with evolving terminology, multi-word entities, noisy/social data, and multilingual coverage

Theme	Supporting Studies	Common Dataset / Data Source	Methods / Approaches	Key Findings / Trends	Limitations / Research Gap
LLM & Retrieval-Augmented Generation (RAG)	Shah & Madiseti (2025); Sekar et al. (2026); Kumar et al. (2026); Barletta et al. (2024)	Dark web content, security documents, vulnerability/CVE-CWE-ATT&CK mappings	Multi-agent LLMs, FAISS/BM25 retrieval + local/API LLMs, LLM-assisted framework mapping	Cuts analyst workload (45 min to 2 min) and factual errors (~38%); improves scraping	7–8s response latency unsuitable for high-frequency monitoring; hallucination risk with incomplete data
ML Anomaly Detection & Classification	Mishra et al. (2022); Al-Shehari et al. (2023); Bavadiya et al. (2025); Dekel et al. (2023)	Public IoT security datasets, CERT insider threat benchmark, Microsoft Malware Dataset	Isolation Forest (IF), hybrid Autoencoder-RF (AE-RF), CNN-LSTM, Multi-Armed Bandit (MAB)	Achieves high detection accuracy (98% IF) and AUC (0.91–0.94); successfully detects polymorphic malware and prioritizes hunting paths	Single-benchmark evaluation; static/offline data only; generalisability to live traffic untested
Knowledge Graphs & GNNs for Correlation & Attribution	Zhou et al. (2024); Ling et al. (2025); Wang et al. (2024); Kumar et al. (2025); Grandhi et al. (2026)	STIX corpora (38k sentences), AnnoCTI benchmark, HoneyPoints/APT-TI-KG, EDR/XDR/DNS telemetry logs	Universal Information Extraction (UIE) + T5, Relational GCN (RGCN), GNNs (GCN, GAT, GraphSAGE), KG mining + RF/GBDT	Accurately reconstructs multi-stage attack paths (96.8% accuracy, 8 ms latency); detects cross-host lateral movement and	High computational costs for multi-hop graph reasoning Degradation in accuracy in the absence of temporal context Vulnerable to graph evasion

Theme	Supporting Studies	Common Dataset / Data Source	Methods / Approaches	Key Findings / Trends	Limitations / Research Gap
				attributes actors early	
Decentralised & Privacy-Preserving CTI Sharing	Jin et al. (2024); Santos et al. (2025); Prajwalasimha et al. (2025); Narayanan & Srinivasan (2025); Kumar et al. (2022); Alaeifar et al. (2024)	Multi-sector sharing platforms, Industrial IoT feeds, simulated SCADA logs, Decentralised IDS (DIDN) traffic	Hybrid on-chain/off-chain in Blockchain, Smart Contracts (PBFT consensus), Federated Learning (FL), Zero-Knowledge Proofs (ZKP), GANs	Removes single points of failure; preserves data privacy and stops poisoning during multi-party CTI dissemination	Consensus latency makes real-time mitigation harder, decentralised pipelines lack automated AI triage and use simulation.
Domain Threat Modelling, Standardisation & Quality Assurance	Chen et al. (2024b); Saulaiman et al. (2025); Erbas et al. (2025); Ali et al. (2025); Avrahami et al. (2025); Sadhwani et al. (2024)	STIX 2.1 JSON feeds, ISO 21434 TARA vehicle assets, Turku maritime testbed, OSINT feeds, Industrial IoT testbeds	STIX graph object filtering, PASTA risk modelling + MISP integration, multi-dimensional quality scoring (accuracy, freshness, relevance)	Standardises threat representation in specialised domains (automotive, maritime, IIoT); eliminates low-quality/redundant IoCs prior to ingestion	Heuristics do not work on all platforms and domain-specific ontologies are still largely ideas that have not been tried in practice.

Table 1. Thematic synthesis / analysis of the included studies.

4.1 Theme 1: NLP & Deep Learning Extraction

The largest and most foundational theme in the reviewed literature focuses on a central challenge in CTI: the sheer volume and unstructured nature of cybersecurity text, including CTI reports, social media posts, and dark-web content, make manual analysis impractical. To address this, researchers increasingly use natural language processing (NLP) and deep-learning models to automatically extract entities, relationships, and threat labels. Chen et al. (2024a) and Sufi (2023), for example, combine NLP with machine-learning clustering or classification to group APT actors and identify threat signals from social media, respectively, while Abdi et al. (2023) use an NLP classification pipeline to automatically label unstructured CTI reports.

More recent studies have shifted toward transformer-based named-entity recognition (NER). Chen et al. (2023) simplify a BERT-CRF model by removing the BiLSTM layer, reducing training time while maintaining 82.6–94.0% accuracy across OSINT and malware corpora. Cheng et al. (2025), meanwhile, combine DeBERTa, BiGRU, and CRF layers with LightGBM to achieve the highest raw entity-extraction accuracy among the reviewed studies. Together, these approaches highlight a recurring accuracy–efficiency trade-off (RQ1): lighter architectures may sacrifice some accuracy in exchange for faster and more practical deployment, whereas more complex hybrid models can achieve higher accuracy at the cost of greater computational requirements.

Several studies also extend beyond basic entity extraction toward more structured forms of CTI. Fieblinger et al. (2024) fine-tune open-source LLMs, including Llama 2 and Mistral, using QLoRA to extract threat triples from raw reports. Huang et al. (2024) combine deep learning with ontology mapping to align extracted information with MITRE ATT&CK techniques, providing a direct link between unstructured text processing and downstream CTI framework integration (RQ4).

Despite this progress, a common limitation remains across the literature (RQ3): model performance is strongly influenced by the quality and completeness of the source data. Models can struggle with evolving threat-actor terminology (Abdi et al., 2023), noisy or unverified social-media content (Sufi, 2023), and limited multilingual coverage. Al-Yasiri

et al. (2024) identify the multilingual issue conceptually, although it has not yet been extensively evaluated empirically. Consequently, extraction accuracy alone does not provide a reliable basis for comparing studies. Differences in corpus quality, language, and source reliability must also be considered when interpreting performance metrics (RQ2).

4.2 Theme 2: LLM & Retrieval-Augmented Generation (RAG)

The most recent studies in the corpus increasingly apply large language models (LLMs), often within retrieval-augmented generation (RAG) architectures, to automate CTI workflows from end to end. Sekar et al. (2026), for example, integrate threat enrichment, API testing, and credential checks into a single RAG platform using BGE embeddings and LLM-based re-ranking. Kumar et al. (2026) combine FAISS and BM25 retrieval with a locally hosted Qwen2.5 model, reporting a 38% reduction in factual errors while reducing analyst workload from approximately 45 minutes to under two minutes per report. Shah and Madiseti (2025) use a multi-agent GPT-4o-mini architecture to scrape, classify, and label dark-web content, showing how LLM-based agents can accelerate both structured and unstructured intelligence gathering. Similarly, Barletta et al. (2024) use an LLM as a fallback mechanism to map CVE and CWE records to the MITRE ATT&CK and ATLAS matrices when rule-based mapping fails.

Together, these studies suggest that LLM/RAG systems are becoming increasingly capable of automating labour-intensive CTI triage and analysis (RQ1), while also providing a direct link between unstructured intelligence and established frameworks such as MITRE ATT&CK (RQ4). However, this approach introduces challenges that are less prominent in graph-based and traditional ML systems (RQ3). Response latency of around 7–8 seconds per query can limit their suitability for high-frequency automated monitoring, while the risk of hallucination remains a significant concern when source information is incomplete or ambiguous. None of the four studies fully resolves these reliability issues. Evaluation is also inconsistent across the theme, with studies using metrics such as retrieval accuracy, expert-rated quality scores, and percentage reductions in errors. This makes direct comparison with the accuracy- and precision-based metrics used in Themes 1 and 2 more difficult (RQ2).

4.3 Theme 3: ML Anomaly Detection & Classification

A smaller but consistent theme in the literature applies conventional and hybrid machine-learning models to anomaly detection across areas such as insider behaviour, malware, and IoT threats. Mishra et al. (2022), for instance, show that standard ML classifiers outperform baseline detection methods on IoT security datasets, while Al-Shehari et al. (2023) report 98% accuracy using an Isolation Forest model to detect rare insider threats on the CERT benchmark. The anomaly-based approach also outperformed supervised alternatives that depend on labelled training examples. Bavadiya et al. (2025) extend this work to zero-day and polymorphic malware detection using hybrid AE-RF and CNN-LSTM architectures, achieving AUC scores of 0.91 and 0.94, respectively, while also providing interpretable detection patterns. Rather than directly classifying threats, Dekel et al. (2023) take a different approach by using a Multi-Armed Bandit model to prioritise the most promising threat-hunting activities.

Overall, these studies demonstrate that ML-based anomaly detection can be highly effective for narrowly defined and well-structured problems (RQ1). However, the evidence is largely limited to controlled evaluation settings. Each study relies on a single benchmark dataset or static file corpus (RQ2), and the authors acknowledge that the models' ability to generalise to live, real-world traffic remains largely untested (RQ3). This limitation mirrors the findings from Themes 1 and 2, suggesting that limited real-world validation is not an issue confined to individual studies but represents a broader gap across the reviewed literature.

4.4 Theme 4: Knowledge Graphs & GNNs for Correlation & Attribution

A number of studies use structured graph representations to model relational and contextual attack semantics (RQ1). Zhou et al. (2024) use 38,121 annotated STIX sentences and a multi-hop reasoning framework with Universal Information Extraction (UIE) and T5 models to find implicit APT associations. They show that relational paths are more useful than traditional graph embedding baselines such as TransE and RotatE. Ling et al. (2025) use the AnnoCTI benchmark and combine LLM semantic prompting with Relational Graph Convolutional Networks (RGCN) to track cross-host lateral moves, achieving 94.8% precision and 96.5% recall. Kumar et al. (2025) also employed Graph Neural Networks (GCN, GAT, and GraphSAGE) on top of network telemetry and claimed that their system achieved a 96.8% success rate in detecting things and a very low latency of 8 ms. As an example of proactive defence, Wang et al. (2024) apply ensemble learning (Random Forest and GBDT) to detect honeypot decoy IPs and link them with threat knowledge graphs for threat actor identification before post-breach forensic logs are generated. Grandhi et al. (2026) apply behavioural graph modelling and Zero-Trust telemetry to detect hidden multi-stage attack paths.

Graph-based methods are better at drawing conclusions, but they have clear structural and operational weaknesses (RQ3). Graph construction pipelines are still very susceptible to gaps in the intelligence feeding into them (Ling et al., 2025). In addition, multi-hop reasoning algorithms are computationally expensive for large enterprise topologies (Zhou et al., 2024) and behaviour graphs are still vulnerable to structural graph evasion and adversarial manipulation (Grandhi et al., 2026). These problems reflect the importance of having flexible graph-learning systems that can deal with large amounts of dynamic streaming data (RQ4).

4.5 Theme 5: Decentralised & Privacy-Preserving CTI Sharing

A key problem for proactive cyber defence is to get organisations to share threat intelligence without revealing private infrastructure or risk data poisoning (RQ3). Jin et al. (2024); Santos et al. (2025) argue that trust boundaries, organisational silos, and platform architectures limit the usefulness of CTI sharing for defence. However, emerging research explores cryptographic and decentralised AI paradigms to secure cross-organisational threat exchanges (RQ1, RQ4). Prajwalasimha et al. (2025) introduce the Federated and Privacy-Preserving Approach (FPPA) that utilises intelligent contracts, Federated Learning (FL), Generative Adversarial Networks (GANs), and Zero-Knowledge Proofs (ZKP) to enable cooperative zero-day threat detection with data privacy preservation. The P2TIF framework was proposed by Kumar et al. (2022) and it utilises deep learning and blockchain to protect the privacy of CTI in Industrial IoT and critical infrastructure. Narayanan and Srinivasan (2025) fuse on-chain and off-chain intelligent contracts with PBFT consensus to share checked IoC hashes and automate decentralised firewall rules, eliminating single points of failure.

But there are significant operational trade-offs with decentralised CTI frameworks (RQ3). Distributed consensus mechanisms can add overhead and delays that make millisecond-level threat triage difficult (Prajwalasimha et al., 2025; Narayanan & Srinivasan, 2025). Furthermore, some of the proposed designs are based on fake industrial or SCADA testbeds instead of real Security Operations Center (SOC) pipelines (RQ2). Thus, scalability and cross-sector AI governance are considered important domains that still need further research (RQ4).

4.6 Theme 6: Domain Threat Modelling, Standardisation & Quality Assurance

Standardising threat schemas and ensuring quality of intelligence taken in are very important for the operational effectiveness of AI in CTI (RQ2, RQ4). Chen et al. (2024b) show that raw CTI feeds often contain noisy, outdated or redundant Indicators of Compromise (IoCs). They remove low-quality IoCs before they impair detection rules later by modelling indicators as STIX graphs. In their 2025 paper, Ali et al. formalise CTI evaluation further by checking feeds for accuracy, completeness, relevance, and freshness. Avrahami et al. (2025) built structured analytics for classifying unstructured Open-Source Intelligence (OSINT) into actionable defensive signals to improve proactive visibility from the outside.

RQ4 In some industrial and cyber-physical domains, researchers use formal threat modelling frameworks for more consistency in risk communication. Saulaiman et al. (2025) map ISO/SAE 21434 Threat Analysis and Risk Assessment (TARA) framework with STIX 2.1 JSON objects to improve the automotive supply chain security. Erbas et al. (2025) use PASTA risk methodology with MISP threat platforms to secure autonomous maritime systems. Barletta et al. (2024) also use matrix alignments and LLM fallbacks to merge CVE and CWE records with MITRE ATT&CK and ATLAS taxonomies. A recurring problem with this theme is that heuristic filtering and cross-domain schemas do not always work well with real-time, multilingual threat streams (RQ3). This points to the need for dynamic, production-grade harmonisation frameworks (Albarrak et al., 2026; Barbieri et al., 2025).

5. Comparative Analysis and Discussion

5.1 Dataset / Data Source Comparison

The use of these disparate datasets has two key implications for model comparability and operational deployability, respectively. Through the works explored, three types of sources for the proposed AI models have commonly appeared: unstructured textual corporas, simulated network telemetry, and active threat feeds.

The use of unstructured text data (e.g., DNRTI, CTI-Reports) is crucial in entity extraction and other NLP related model architectures due to the need for vast amounts of text data (DNRTI contains more than 182,000 words that map to 13 different entities) to adequately train transformer(X) models. Similarly, many studies use the MITRE ATT&CK and ATLAS frameworks to anchor behavioral knowledge of the extracted threat entities. Though the use of these structured/semi-structured sources are paramount in the training and calibration of extraction models, these sources lack dynamism like active attack profiles.

Conversely, these active/interactive data sources are common in models that aim to forecast attacks early, predict an attack's eventual destination, or tie data to a specific actor (honeypot captured IP addresses, Dark Web forums like RaidForums and CODA database, VirusTotal). The drawback to utilizing these live feeds is that each type is exceptionally noisy, highly transient, and class imbalances were often extreme so the AI might achieve greater precision over the aforementioned methods, though less in practical applications and operational versatility on the former benchmarks.

5.2 Method / Approach Comparison

According to the literature review, we have seen a clear evolution from manually analysing threat intelligence to automating the extraction and reasoning process through techniques such as deep learning, GNNs, and LLMs.

Transformer-based architectures have become predominant in extracting information from unstructured reports. Combinations of DeBERTa or BERT, with Conditional Random Fields (e.g., BERT-CRF, DeBERTa-BiGRU-CRF) show great potential in Named Entity Recognition (NER) with a good combination of high accuracy and reasonable computation cost. However, these general NLP based models process information in isolation.

To detect multi-stage threats and attributions, there are several approaches based on graph-based methods such as Knowledge Graphs (KGs) and Graph Neural Networks (GNNs) (e.g. GCN, GAT, GraphSAGE). Graph-based methods are highly effective in linking dispersed indicators of compromise and revealing the hidden lateral movements with high accuracy and very low latency (e.g. 8ms for GNN prediction).

Recently, RAG and multi-agent LLMs such as GPT-4o-mini, and Qwen2.5 have become popular for enabling automated analytical workflows and reporting the threat intelligence in human-readable documents. LLMs are strong for qualitative summary and zero-shot reasoning but can suffer from hallucinations if the source information is sparse. Therefore, even though graph-based approaches could be a faster and less error-prone solution for detection, hybrid RAG-LLMs approaches seem most applicable in automating repetitive, dull tasks in collecting and analysing threat intelligence.

5.3 Security Functionality / Process Comparison

The security functionalities of the models observed range based on the nature of the underpinning AI models. The systems relying on NLP & Transformer based models such as BERT-CRF act as Extraction pipelines. They consume the raw text-based reports and process through NER to extract structured threat triples such as Tactics, Techniques and Procedures (TTPs) or Indicators of Compromise (IoCs).

Graph Neural Network (GNN) or KG methodologies instead primarily operate as a "correlation and reasoning engine". They can be seen as mapping actors (like Users, IPs) as nodes and activity as a directed edge in order to automatically detect anomalies, to compute dynamic risk scores and potentially isolate entities during multi-stage attacks, for example. Newer Retrieval-Augmented Generation (RAG) and multi-agent LLM-based systems can also be considered as "automated analysts". These systems ingest questions, search for relevant information stored in vector databases like FAISS and to provide extensive and readable threat reports which largely automate the last part of the CTI process.

5.4 Evaluation Metrics and Results

The evaluation metrics among studies vary considerably depending on the CTI task and therefore a meaningful cross-comparison between studies is difficult to conduct. NLP and extraction studies mainly use standard classification performance measures of accuracy, precision, recall and F1 score, along with text specialized metrics of ROUGE score or NER accuracy.

Graph based studies and attacker-attribution studies most commonly publish Macro-F1, Micro-F1 and Hits@k due to high imbalances in class distribution in threat data. The GNN model achieves excellent real-time performance by achieving a high accuracy up to 96.8% while having a ridiculously low detection latency of only 8ms. On the other hand, those who are assessing an LLM or RAG based model look for: Retriever Accuracy, Expert quality score and response time (latency). However, LLMs increase the efficiency for an analyst so rapidly it compromises performance time (average 7 - 8.4 sec), it thus potentially impacts its usability for rapid and automated network surveillance.

5.5 Strengths and Limitations

Across the analysed body of work, one recurring positive theme is AI's potential for rapidly mining large amounts of unstructured text that human analysts could not otherwise hope to cover. Graph-based methods have been particularly adept at precisely detecting hidden lateral movement paths with very low latency, and LLMs are proving themselves uniquely capable of qualitative synthesis and zero-shot reasoning on top of CTI report.

However, three interrelated limitations frequently surface across the analysed reports. First, AI extractive models are inherently bound by the amount of input data that's available to the model; if source documents lack the relevant historical context for the target individual or actor. Second, high-performance LLM and RAG based AI have the trade-offs of significant computational costs and response lag times, and a chronic liability for hallucination when processing incomplete intelligence. Third, almost all evaluated systems were on static and or offline benchmark / simulation training data and were not evaluated on the demands that a real-world high throughput system faces.

6. Research Trends and Challenges

Analysis of the 40 reviewed studies published between 2022 and 2026 reveals a clear methodological evolution alongside persistent operational challenges in AI-driven Cyber Threat Intelligence (CTI).

Emerging Research Trends (2022–2026)

6.1 From Static Text Extraction to Semantic Graph Reasoning

Early studies mainly focused on isolated Named Entity Recognition (NER) and machine learning classifiers on static text (Mishra et al., 2022; Chen et al., 2023). Recent work from 2024 to 2026 has focused on graph neural networks (GNNs), knowledge graph (KG) triple extraction, and relational reasoning to uncover multi-stage attack paths and cross-host lateral movements (Zhou et al., 2024; Ling et al., 2025; Kumar et al., 2025).

6.2 Multi-Agent LLMs and Hybrid RAG Pipelines Adoption

In recent years, studies have focused on automating report generation, speeding up dark web scraping, and drastically reducing triage workloads for analysts by applying large language models (LLMs) with Retrieval-Augmented Generation (RAG) and dense vector retrievers (FAISS, BM25) (Shah & Madiseti, 2025; Kumar et al., 2026; Sekar et al., 2026) between 2024 and 2026.

6.3 Decentralised and Privacy-Preserving Collaborative Defences

To address risks associated with single point failures and data exposure in centralised sharing hubs, recent architectures use hybrid blockchain smart contract, federated learning (FL), and zero-knowledge proofs (ZKP) to securely exchange verified Indicators of Compromise (IoCs) across multi-sector boundaries (Kumar et al., 2022; Narayanan & Srinivasan, 2025; Prajwalasimha et al., 2025).

6.4 Harmonisation with Standardised Threat Ontologies

Recent literature indicates improved concordance with standardised taxonomies (e.g., STIX 2.1, MITRE ATT&CK, MITRE ATLAS) and cross-domain risk frameworks (ISO 21434 TARA, PASTA) to structure unstructured intelligence from automotive, maritime and Industrial IoT sectors (Barletta et al., 2024; Erbas et al., 2025; Saulaiman et al., 2025).

Persistent Challenges

1) The Accuracy–Latency Dilemma:

High-accuracy generative AI and multi-agent LLM systems have high response latencies (7- 8.4 seconds) and are not practical for millisecond-level, real-time threat interception (Shah & Madisetti, 2025; Sekar et al., 2026; Kumar et al., 2026). In contrast, ultra-fast graph architectures (e.g. 8 ms latency in Kumar et al., 2025) do not possess natural language synthesis capabilities.

2) Vulnerability to Hallucinations and Incomplete Metadata:

Nevertheless, LLM-based reasoning is still prone to factual hallucination and classification degradation in the presence of fragmented threat feeds, chronological gaps, or novel attacker terminology (Abdi et al., 2023; Barletta et al., 2024; Ling et al., 2025).

3) Limited Live, Production-Ready Network Testing:

Most published models are only validated in isolation via static benchmarks (e.g., DNRTI, CERT insider threat data) or in controlled testbeds, and their generalisability against dynamic, adversarial evasion in high-throughput live network environments remains largely unverified (Al-Shehari et al., 2023; Bavadiya et al., 2025; Grandhi et al., 2026).

7. Research Gaps and Future Directions

7.1 Research Gap 1

Many studies evaluated AI-driven threat extraction and detection frameworks on static, historical datasets or stand-alone simulation environments, which leaves a lack of clarity on the applicability of reported high-accuracy performance to ongoing, high-volume network streams and active adversarial evasion in real-world settings (Al-Shehari et al., 2023; Bavadiya et al., 2025; Cheng et al., 2025; Grandhi et al., 2026; Narayanan & Srinivasan, 2025).

Future Direction

Future work should consider the deployment of AI threat analysis frameworks in operational Security Operations Center (SOC) environments with live streaming telemetry, focusing on the scalability of reporting throughput, reduction of operational false positives, and the decline of models in the presence of dynamic concept drift.

7.2 Research Gap 2

Generative AI and retrieval-augmented generation (RAG) systems greatly reduce the analyst triage time but exhibit significant response latencies (7–8.4 seconds) and ongoing hallucination risks due to incomplete data. Ultra-rapid graph neural networks do not possess the natural language generation and contextual analysis capabilities necessary for security personnel (Barletta et al., 2024; Fieblinger et al., 2024; Kumar et al., 2025, 2026; Sekar et al., 2026; Shah & Madisetti, 2025).

Future Direction

Future work should focus on the development of dual-layer hybrid architectures integrating ultra-low latency GNNs for instant threat filtering and asynchronous, locally quantised LLMs under strict retrieval validation to generate hallucination-resistant CTI reports.

7.3 Research Gap 3

Current decentralised and collaborative threat sharing frameworks rely heavily on blockchain consensus and cryptographic proof mechanisms. Yet, these frameworks suffer from significant computational overheads and latency delays, which preclude their real-world deployment for millisecond-level threat mitigation in critical infrastructure (Jin et al., 2024; Kumar et al., 2022; Narayanan & Srinivasan, 2025; Prajwalasimha et al., 2025; Santos et al., 2025).

Future Direction:

Future work should focus on the development of lightweight consensus protocols, off-chain state channels, and hardware-accelerated zero-knowledge computing specifically for CTI exchange, augmented with automated AI triage nodes pre-filtering Indicators of Compromise prior to on-chain execution.

7.4 Research Gap 4

Current CTI extraction pipelines usually work on English-language OSINT and often process raw intelligence without an automatic quality check, leading to many errors and alert fatigue caused by noisy, redundant, or old indicators (Abdi et al., 2023; Albarrak et al., 2026; Ali et al., 2025; Al-Yasiri et al., 2024; Chen et al., 2024b).

Future Direction:

Researchers should develop automated quality-assurance systems for threat feeds that assess them against a set of criteria such as accuracy, completeness, timeliness and relevance. Further, the multilingual NLP encoders need to be improved to process unstructured threat data from multiple sources like dark web and global web.

8. Conclusion

The majority of the 40 publications, analysed in this systematic literature review, were published between the years 2022 and 2026. The main applications of Artificial Intelligence (AI) for Cyber Threat Intelligence (CTI) are dynamic semantic graph reasoning by Graph Neural Networks (GNNs), automated threat triage, information distillation by Large Language Models (LLMs), and retrieval-augmented generation (RAG), which are different from the old ways based on static Natural Language Processing (NLP) based text extraction. The literature also showed that AI is slowly finding its way into decentralised and private CTI sharing systems. Intelligence is also being translated into standardised threat ontologies. The results show that standard artificial intelligence systems can assist human analysts by modelling complex attack vectors and standardising the way in which threats are defined across different domains of expertise.

Despite the big steps forward, the main problem of integrating AI in CTI is that accuracy-latency problems are still very common, and a lot of work is still being done on static, offline benchmark datasets. GNNs have no latency time for real time detection, while state of the art LLMs and RAG take too long to process results and may have factual hallucinations if they do not get all the information they need. More importantly, the models have not been tested sufficiently in real world high-throughput networks to understand their effectiveness in different situations or against complex adversarial evasion schemes. In future, researchers will have to go beyond the current offline, computer generated benchmarks and explore real-life Security Operations Center (SOC) operations to identify the best dual-tier, hybrid models, lightweight consensus protocol, automated quality assurance system and AI that can identify threats in real time and make the analysts' job easier.

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Group Contribution Table

No.	Student Name	Student ID	Main Responsibility	Section(s) Completed	Evidence / Notes
1	MUHAMMAD ZAINUL AKHTAR BIN NOR AZMAN	52215225187	Group Leader, Report Compilation and overall editing.	Section 6 - Research Trends and Challenges, Section 7 - Research Gaps and Future Directions	Completed
2	ARIFF IMRAN BIN ALWI	52215225001	Thematic literature synthesis and results analysis; grouped the 40 reviewed papers into six themes and mapped findings to RQ1–RQ4	Section 1, Introduction Section 2 - Review Questions Section 4 – Results and Thematic Synthesis (incl. Table 1: Thematic Synthesis Summary)	Completed
3	MUHAMMAD FIRDAUS BIN HAIRUMAINI	52215225004	Designed the SLR methodology protocol, constructed the search strategy and eligibility criteria, and formulated the PRISMA 2020 screening flow diagram.	Section 3 – SLR Methodology (incl. Section 3.1 Search Strategy, Section 3.2 Inclusion/Exclusion Criteria, Section 3.3 Study Selection Process & PRISMA 2020 Flow Diagram, and Section 3.4 Data Extraction)	Completed

No.	Student Name	Student ID	Main Responsibility	Section(s) Completed	Evidence / Notes
4	MUHAMMAD ISYRAF BIN ISMAIL	52215124383	Conducted the comparative analysis of the reviewed studies and drafted the final conclusion.	Section 5 – Comparative Analysis and Discussion (incl. dataset, method, and metrics comparison) Section 8 – Conclusion	Completed

Final Submission Check

- PRISMA diagram is placed in Section 3.3 and uses the group's actual screening numbers.
- The Thematic Synthesis / Analysis Table is completed and placed before the detailed theme discussions.
- A minimum of 3 and maximum of 6 themes are discussed.
- At least TWO review questions are included.
- The discussion synthesises several studies rather than describing papers one by one.
- Research gaps are supported by evidence from multiple studies.
- Future directions respond directly to the identified research gaps.
- APA 7th edition is used consistently.
- The Group Contribution Table is completed after References.

ASSESSMENT RUBRIC - ASSIGNMENT 1 SECTION B

Systematic Literature Review (SLR) - 30%

Marking guidance: Award marks within each criterion based on the quality of evidence demonstrated in the submitted SLR. The descriptors below indicate the expected level of performance.

Assessment Criterion	Max Marks	Excellent	Good	Satisfactory	Needs Improvement	Marks Awarded
1. Introduction & Review Questions	3	Clear, focused introduction establishes the cybersecurity context, significance, scope and purpose. At least 2 well-aligned review questions are specific and answerable from the selected literature.	Clear introduction and relevant review questions, with minor gaps in focus, scope or alignment.	Basic introduction and review questions are present but broad, descriptive or only partly aligned with the SLR.	Introduction is unclear or lacks purpose. Review questions are missing, weak, not answerable from the literature, or unrelated to the assigned topic.	____ / 3
2. SLR Methodology & PRISMA	4	Search strategy, eligibility criteria, selection process and data extraction are clearly reported. PRISMA diagram is correctly placed and uses actual, internally consistent screening numbers and exclusion reasons.	Method is mostly complete and PRISMA process is clear, with only minor omissions or inconsistencies.	Method is described at a basic level. PRISMA is included but some numbers, criteria or process details are unclear.	Method is incomplete or difficult to reproduce. PRISMA is missing, incorrectly placed, uses invented/inconsistent numbers, or does not reflect the actual search process.	____ / 4
3. Thematic Literature Synthesis	6	Studies are organised into 3-6 meaningful themes. Each theme synthesises multiple studies, identifies patterns and contrasts, and uses the analysis table effectively to support the written discussion.	Themes are relevant and generally well synthesised, though some discussion remains descriptive or uneven across themes.	Themes are identifiable but synthesis is limited. Several paragraphs summarise studies individually rather than integrating evidence.	Themes are poorly defined, unsupported, or structured paper-by-paper. Analysis table is incomplete or not connected to the discussion.	____ / 6
4. Comparative Analysis & Discussion	5	Critically compares datasets/data sources, methods, processes, evaluation metrics, results, strengths and limitations. Explains why differences matter and avoids unsupported claims about a single "best" method.	Good comparison across most required components, with some critical interpretation and appropriate evidence.	Basic comparison is present but relies heavily on listing information with limited explanation of significance or trade-offs.	Little meaningful comparison. Discussion is mostly descriptive, fragmented, inaccurate, or unsupported by the reviewed literature.	____ / 5
5. Research Trends, Gaps, Future Directions & Conclusion	5	Identifies well-supported trends and recurring challenges. Research gaps are derived from multiple studies and future directions respond directly to those gaps. Conclusion synthesises the main findings without introducing new evidence.	Relevant trends, gaps and future directions are identified and generally supported, with minor weaknesses in linkage or depth.	Some trends and gaps are stated but evidence is limited, generic or based on isolated studies. Future directions are broad.	Gaps are vague, unsupported or confused with single-study limitations. Future directions are unrelated to the evidence. Conclusion is weak or introduces new material.	____ / 5
6. Academic Writing, Evidence & APA 7th	3	Writing is coherent, analytical and academically appropriate. Claims are well supported. APA 7th in-text citations and reference list are accurate and consistent, with strong alignment between cited sources and references.	Generally clear academic writing with appropriate evidence and only minor APA or organisation errors.	Meaning is understandable but writing is repetitive, descriptive or inconsistently referenced. Several APA errors are present.	Writing lacks clarity or academic structure. Evidence is weak, citations/references are substantially incorrect, or sources are missing/inconsistent.	____ / 3

Assessment Criterion	Max Marks	Excellent	Good	Satisfactory	Needs Improvement	Marks Awarded
7. Generative AI Use & Academic Integrity	2	AI use is transparently declared in the Generative AI Usage Acknowledgment. AI is used as a support tool rather than a substitute for analysis. Output is critically checked, rewritten where needed, and all facts/citations are verified. No fabricated references.	AI use is declared and generally responsible. Most generated content is checked and integrated appropriately, with minor evidence of over-reliance or insufficient revision.	AI use is declared but substantial wording or analysis appears insufficiently verified, critically revised or integrated with the actual reviewed literature.	AI use is undeclared or inappropriate; the acknowledgement is incomplete; generated material is submitted with little critical review; or fabricated/inaccurate citations, references or evidence are present.	____ / 2
8. Teamwork & Group Contribution	2	Work is distributed fairly and contributions are clearly documented. Sections are integrated into one coherent paper, demonstrating effective coordination, peer review and shared responsibility.	Most members contribute meaningfully and responsibilities are documented; minor imbalance or integration issues are evident.	Contribution is uneven or weakly documented. The paper shows noticeable differences in quality, style or coordination between sections.	Significant imbalance or lack of participation is evident. Contribution table is incomplete, responsibilities are unclear, or the submission shows little evidence of collaboration.	____ / 2
TOTAL MARKS (/30)	30					_____

Area of improvement from lecturer:

Generative AI Content Use Level Guide

This criterion assesses responsible and transparent use based on the submitted AI Usage Acknowledgment and the quality of the work. It is not based on an automated AI detector.

Level	Description	Expected Student Practice	Indicative Mark
Level 3 - Responsible / Appropriate	AI supports the work but does not replace the student's analysis.	Declare the tool and prompts; verify claims and references; rewrite and integrate output; demonstrate original synthesis from the 40-50 papers.	2.0
Level 2 - Acceptable with Minor Concerns	AI use is declared and mostly appropriate, but some content shows limited critical revision or verification.	Improve checking, paraphrasing, source verification and connection to actual reviewed studies.	1.0-1.5
Level 1 - High Dependence	Large portions appear AI-generated or formulaic, with limited evidence of student analysis or source-based verification.	Substantially revise using actual evidence; verify all citations and claims; strengthen original interpretation.	0.5
Level 0 - Unacceptable / Undeclared	AI use is not declared, or generated output contains fabricated evidence/citations, or replaces the required student analysis.	Does not meet academic integrity and assignment expectations.	0

Teamwork evidence: The marker may refer to the Group Contribution Table, consistency of the integrated paper, and other contribution evidence required by the lecturer. A contribution table alone does not automatically demonstrate equal teamwork.